ASSIGNMENT-3-CO3

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COURSE	Mobile Computing
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Q1. Wireless Network Architecture and Mobility Analysis

Part A – Introduction to Wireless Networks

Shared work. A wireless network connects mobile nodes (phones, laptops, vehicles) to each other and to a fixed network over radio links instead of cables. Its main components are:

Component	Role
Mobile node (MN)	End device that moves and communicates over the air
Base station / access point (AP)	Radio gateway that serves a coverage area (cell) and relays traffic to the wired side
Wireless link	Shared radio channel between MN and AP; limited bandwidth, error-prone
Backbone / wired network	Switches and routers connecting APs to servers and the Internet

Gateway / server	Connects to external networks and hosts the application

Architecture: MN ⇄ (wireless link) ⇄ AP/Base station ⇄ (wired backbone) ⇄ Router/Gateway ⇄ Internet / Server. Constraints are low and shared bandwidth, intermittent connectivity (fading, interference, shadowing), limited battery, and mobility between cells.

Student A – Network Architecture Analysis.

- Components and roles are as in the table above; each AP covers one cell and all MN traffic passes through it.
- Wireless characteristics: broadcast medium, higher bit-error rate than wire (10^{-3}–10^{-6} vs about 10^{-9}), variable signal strength with distance, interference and collisions, half-duplex operation in many systems.
- Limited bandwidth means many users share one channel, so throughput per user drops as more MNs attach and retransmissions waste capacity.
- Connectivity is lost when the MN leaves coverage or signal falls below threshold, so sessions need reconnection or handoff.
- Communication path: the MN sends a frame to its AP; the AP forwards it over the backbone to the destination; the reply returns the same way. Two MNs in the same network communicate via the AP in infrastructure mode.

Student B – Mobility and Performance Analysis.

Representative mobility scenarios: (1) stationary user, (2) pedestrian at about 1.5 m/s, (3) vehicle at about 15–30 m/s moving across several cells.

Scenario	Handoffs per minute	Interruption	Packet loss	Delay
Stationary	0	None	Very low (channel errors only)	Stable, low
Pedestrian	Low	Short, occasional	Low	Small spikes at handoff
Vehicle	High	Frequent	Noticeably higher	Large spikes; jitter

- Handoff and interruption: during a handoff the MN detaches from the old AP and attaches to the new one; packets sent in this gap are lost or delayed (typically tens to hundreds of milliseconds).
- Packet loss comes from signal fading, collisions and handoff gaps; delay comes from retransmissions, queueing at the AP and re-association.
- Changing conditions (more users, weaker signal, higher speed) raise loss and delay and reduce throughput. Higher mobility gives more handoffs and therefore lower communication efficiency.

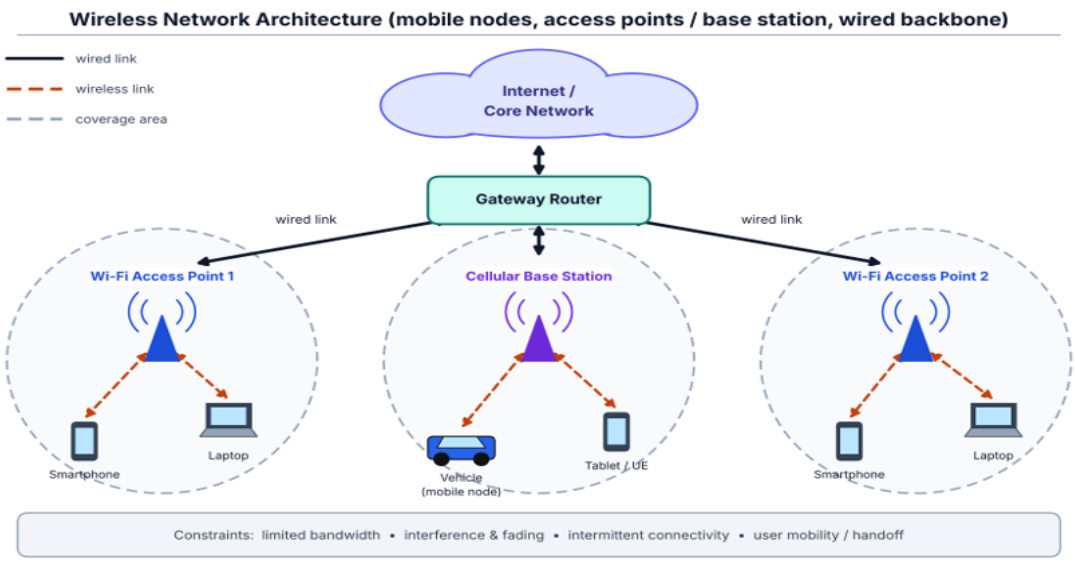


Figure 1: Wireless network architecture

Part B – Wireless Mobility Scenario

Shared scenario. A user on a video call moves from the coverage of AP1 to the coverage of AP2 while the session is active.

Student A – Analytical investigation.

- Successful communication: the MN is inside a cell, signal ≥ threshold, packets are acknowledged and delivered in order.
- Interrupted communication: signal drops below threshold, the MN is between cells, or re-association with AP2 is incomplete.
- Sequence of events: (1) MN connected to AP1 → (2) signal from AP1 weakens → (3) MN scans and detects AP2 → (4) MN disassociates from AP1 → (5) MN associates with AP2 → (6) traffic resumes through AP2.

- Illustrative sample record (assumed values for the worked example): normal delay 40 ms and 0 % loss; during handoff delay about 300 ms and about 8 % loss for roughly 0.3 s; after handoff delay returns to 40 ms.
- Increasing mobility (more handoffs per minute) multiplies these interruptions, so average delay and loss both rise.

Student B – Mobility logic.

```
ALGORITHM HandoffDecision(MN, CurrentAP, Threshold)
INPUT: MN – mobile node; CurrentAP – serving AP; Threshold – minimum signal strength
OUTPUT: Serving AP after the decision
BEGIN
  WHILE session is ACTIVE DO
    Signal ← SignalStrength(MN, CurrentAP)
    IF Signal < Threshold THEN
      Candidate ← BestAP(ScanNeighbors(MN))
      IF Candidate exists AND SignalStrength(MN, Candidate) > Signal THEN
        DISASSOCIATE from CurrentAP
        ASSOCIATE with Candidate
        CurrentAP ← Candidate
      ELSE
        MARK connectivity LOST and RETRY after timer
      END IF
    END IF
  END WHILE
```

RETURN CurrentAP

END

- The MN changes its point of attachment when the serving signal falls below the threshold and a stronger AP exists.
- Affected conditions: delay, packet loss and throughput during the switch, and any session bound to the old address.
- Termination: the scenario ends when the session completes or the MN stays out of all coverage beyond the retry timeout.
- Repeated mobility lowers efficiency because every handoff adds re-association time and retransmissions.

Q2. Mobile IP and Mobility Management

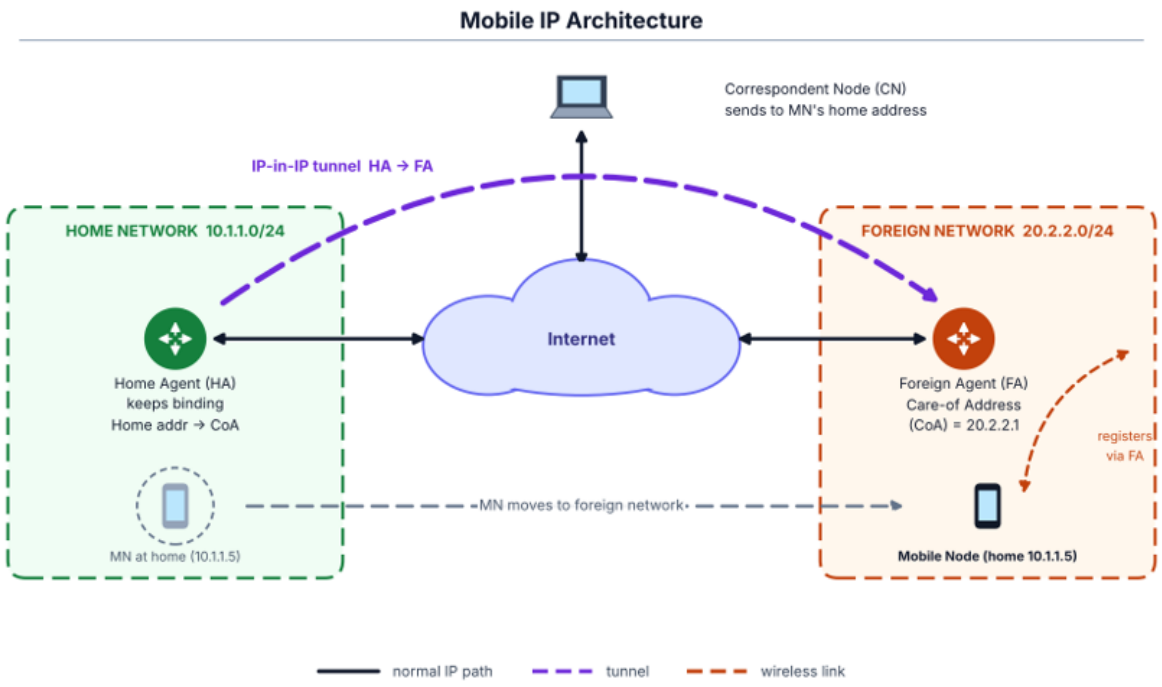
Shared work. Mobile IP lets a mobile node keep one permanent home address while it moves between networks, so that TCP/UDP sessions survive a change of network.

Architecture and terminology

Entity	Meaning
Mobile Node (MN)	Device that changes its point of attachment but keeps its home address
Home Agent (HA)	Router on the home network that keeps the MN's binding (home address → care-of address) and tunnels packets to it
Foreign Agent (FA)	Router on the visited network that provides a care-of

	address and delivers decapsulated packets to the MN
Correspondent Node (CN)	The host communicating with the MN; it only knows the home address
Home address	Permanent IP address of the MN
Care-of address (CoA)	Temporary address that identifies the MN's current location

Topology: CN → Internet → HA (home network) ═ tunnel ═► FA (foreign network) → MN.



Agent discovery and registration

1. **Agent discovery.** HA and FA broadcast Agent Advertisements periodically (ICMP router advertisement extension). The MN compares the network prefix to learn whether it is at home or visiting. If no advertisement arrives, the MN sends an Agent Solicitation.

2. **CoA acquisition.** The MN gets a foreign-agent CoA (the FA's address) or a co-located CoA (an address obtained by DHCP).
3. **Registration Request.** MN → FA → HA carrying home address, CoA, lifetime, identification field and authentication extension.
4. **Binding.** The HA validates the request, stores Binding[HomeAddress] = CoA, and sends a Registration Reply (accepted or denied) back via the FA to the MN.
5. **Re-registration.** The MN registers again before the lifetime expires or when it moves; on returning home it deregisters.

Packet delivery and tunnelling

1. The CN sends a packet to the MN's home address; normal routing delivers it to the home network.
2. The HA intercepts it (proxy ARP) and looks up the binding.
3. The HA encapsulates the packet in a new IP header (IP-in-IP) addressed to the CoA and tunnels it.
4. The FA (or the MN, if co-located) decapsulates and delivers the original packet to the MN.
5. In the reverse direction the MN sends packets directly to the CN with its home address as source (via the FA as default router). This asymmetry is triangular routing.

Worked example (illustrative addresses): MN home address 10.1.1.5, HA 10.1.1.1, FA/CoA 172.16.2.1, CN 192.168.5.9. CN sends 192.168.5.9 → 10.1.1.5; HA wraps it as 10.1.1.1 → 172.16.2.1; FA unwraps and forwards to 10.1.1.5 on the foreign link; the reply goes 10.1.1.5 → 192.168.5.9 directly.

Student A – Mobile IP operation

- **Mobile Node:** detects movement from agent advertisements, obtains a CoA, registers it with the HA and keeps using its home address for applications.
- **Home Agent:** maintains the mobility binding table, intercepts packets for the MN's home address, tunnels them to the CoA, and authenticates registrations.
- **Foreign Agent:** advertises itself, relays registration messages, decapsulates tunnelled packets and delivers them over the local link.
- **Correspondent Node:** sends packets to the home address without knowing the MN has moved (in base Mobile IPv4).
- **Registration and location update:** MN sends a Registration Request each time its CoA changes; the HA updates the binding and replies; the binding expires when its lifetime ends.
- **Delivery and tunnelling:** as in the five-step flow above.

```
ALGORITHM TunnelPacketToMobileNode(Packet, DestinationHomeAddress)
INPUT: Packet – original IP packet; DestinationHomeAddress – MN home address
OUTPUT: DELIVERY_SUCCESS or DELIVERY_FAILURE
BEGIN
  IF Binding[DestinationHomeAddress] does not exist THEN
    DROP Packet
    RETURN DELIVERY_FAILURE
  END IF
  CoA ← Binding[DestinationHomeAddress]
  EncapsulatedPacket ← ENCAPSULATE(Packet, CoA)
```

```
  FORWARD EncapsulatedPacket TO CoA
  IF TunnelDeliverySuccessful THEN
    DELIVER original Packet TO MobileNode
    RETURN DELIVERY_SUCCESS
  ELSE
    RETURN DELIVERY_FAILURE
  END IF
END
```

Student B – Mobile IP challenge analysis

Challenge	Analysis
Registration delay	Each move needs agent discovery plus a round trip MN–FA–HA–FA–MN. If the HA is far away this takes hundreds of milliseconds, during which packets go to the old CoA and are lost.
Tunnelling overhead	IP-in-IP adds a 20-byte header per packet (about 4 % on a 500-byte packet, about 40 % on a 50-byte voice packet) and extra encapsulation processing at HA and FA.
Triangular routing	CN → HA → FA → MN but MN → CN directly. The forward path can be much longer than the direct path, adding delay and load on the home network.

Packet loss during mobility	Packets in flight to the old CoA are dropped between detachment and registration; TCP may treat this as congestion and shrink its window.
Security	Registration messages can be spoofed to redirect traffic (redirection attack), so MN–HA authentication (shared secret, keyed MD5/HMAC), replay protection via the identification field and tunnel protection are required.
Scalability	The HA is a single point of failure and a bottleneck: it handles every registration and every tunnelled packet for all its MNs. Frequent movers generate heavy signalling.

Joint design question

If a mobile user frequently changes its network of attachment during an active session, what problems can occur in Mobile IP and how can they be reduced?

Problems: repeated registration delay and signalling, bursts of packet loss at each move, long triangular paths through a distant HA, HA overload, TCP throughput collapse after losses, and exposure of mobility signalling to attack.

Ways to reduce them:

- **Route optimisation:** the HA informs the CN of the CoA (binding update) so the CN sends directly, removing the triangular path.
- **Smooth handoff:** the old FA buffers packets and forwards them to the new FA (previous-FA notification), so in-flight packets are not lost.

- **Hierarchical Mobile IP / regional registration:** register locally with a nearby Mobility Anchor or regional FA while the HA is told only about macro moves; this cuts the delay to the distant HA.
- **Fast handover:** predict the next network from signal strength and prepare the CoA before detaching.
- **Co-located CoA or Mobile IPv6** to avoid FA dependence and remove tunnelling overhead through route optimisation.
- **Authentication and lifetimes:** keep strong authentication but use sensible binding lifetimes to limit signalling.

Justification: the delay and loss are dominated by the round trip to the HA and by the detach-to-register gap; moving registration closer to the MN (hierarchical) and buffering during that gap (smooth handoff) attack both causes directly, while route optimisation removes the extra path.

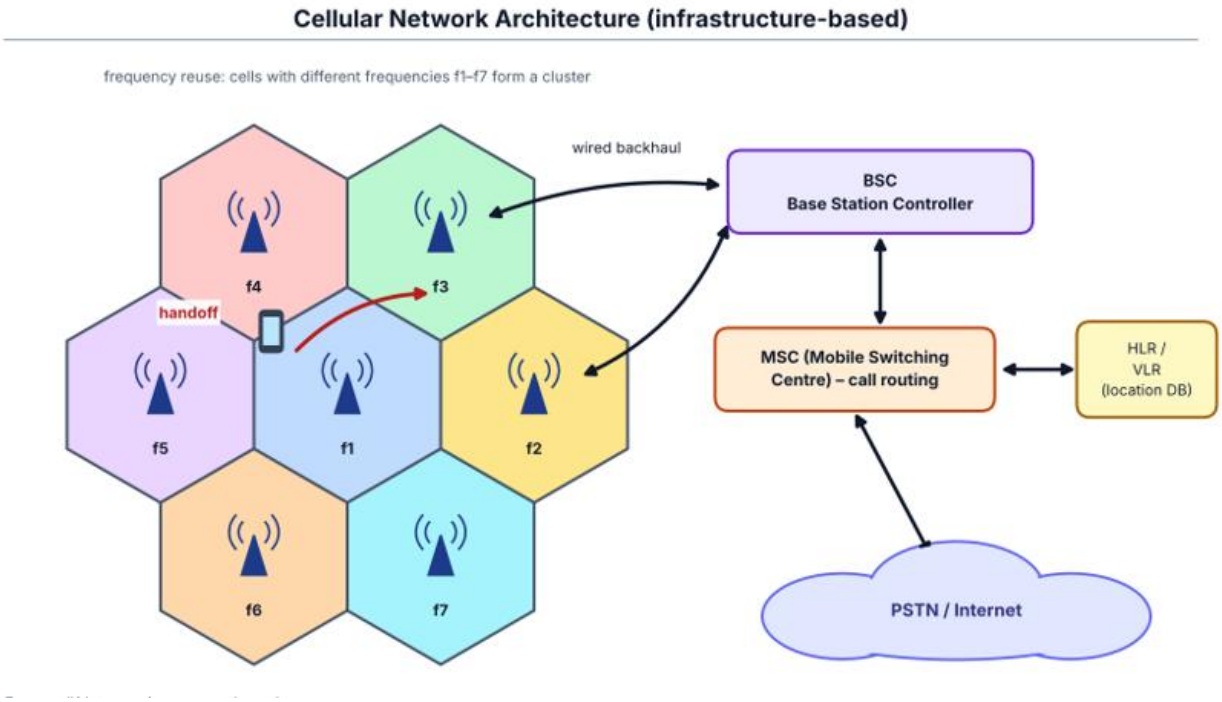
Cross-understanding. Student A can explain the challenges above (delay, overhead, triangular routing, loss, security, scalability); Student B can explain the full discovery → registration → tunnelling flow described in this question.

Q3. Cellular Networks versus Ad Hoc Wireless Networks

Shared work. A cellular network is infrastructure-based: the area is divided into cells, each served by a base station connected to a central switching core. An ad hoc network is infrastructure-less: mobile nodes form a temporary network among themselves and forward each other's packets over multiple hops.

Architecture sketch:

- Cellular: MN ⇄ Base Station (cell) ⇄ Mobile Switching Centre / Core ⇄ other networks. All traffic goes through the base station and core.
- Ad hoc: MN1 ⇄ MN2 ⇄ MN3 ⇄ MN4. Every node is both host and router; links appear and disappear as nodes move.



Student A – Cellular network investigation

- **Architecture:** cells (often hexagonal), base stations with antennas, base station controllers, a mobile switching centre and gateways; frequencies are reused in non-adjacent cells.
- **Role of base stations and infrastructure:** base stations provide radio access, channel allocation and power control; the switching centre handles call setup, routing, authentication and billing.

- **Centralised communication:** even two phones in the same cell communicate through the base station and core, which gives central control and quality of service.
- **Handoff between cells:** when signal falls below a threshold the call moves to a neighbouring base station (hard handoff: break-before-make; soft handoff: make-before-break). The core coordinates it, so calls continue while moving.
- **Scalability and coverage:** capacity grows by cell splitting, sectoring and frequency reuse; coverage is wide and planned, but needs expensive towers and spectrum.
- **Suitable applications:** voice calls, mobile broadband, SMS, mobile banking, video streaming, wide-area IoT.

Student B – Ad hoc network investigation

- **Architecture:** a set of mobile nodes with wireless interfaces and no base station; nodes within radio range are neighbours.
- **Communication between nodes:** direct if in range, otherwise through intermediate nodes chosen by a routing protocol.
- **Dynamic topology:** movement, battery depletion and interference change links continuously, so routes break and must be rediscovered.
- **Multi-hop communication:** a packet from S to D travels S → N1 → N2 → D; each node forwards, so delay and loss grow with hop count and bandwidth is shared.
- **Absence of fixed infrastructure:** no central controller, no planning; the network is set up quickly and is self-organising.
- **Suitable applications:** disaster relief and rescue, military battlefield communication, sensor networks, vehicular (VANET) communication, temporary meetings and events.

Comparative analysis

Parameter	Cellular Network	Ad Hoc Network
Infrastructure	Fixed base stations, switching centre and backbone	None; nodes are the network
Communication	Single hop to the base station, then via the core (centralised)	Multi-hop, peer to peer (distributed)
Routing	Routing is done by fixed infrastructure; the MN needs none	Every node routes; needs protocols such as DSDV, AODV, DSR
Mobility	Mobility is managed centrally (location registers, handoff)	Nodes move freely; mobility changes the topology itself
Topology	Static (cells fixed), star-like per cell	Dynamic, arbitrary, changing graph
Scalability	Scales well by adding cells and reusing frequencies	Degrades with many nodes and hops (overhead, shared channel)
Handoff	Network-controlled handoff between base stations	No handoff; routes are repaired or rediscovered
Failure handling	Base station failure drops its cell, but redundancy and neighbouring cells help	A node failure only breaks routes through it; alternate routes are found
Applications	Voice, mobile data, mass-market	Disaster, military, sensor and

	services	vehicular networks, temporary networks

Joint engineering decision

For an emergency or disaster environment where fixed infrastructure may be unavailable, which architecture would you select and why?

I would select an **ad hoc network** (a mobile ad hoc network, MANET). Disasters often destroy towers and power, and a cellular network cannot work without them. An ad hoc network needs no infrastructure, can be deployed in minutes, is self-healing because nodes find new routes when others fail, and lets rescue teams talk directly or through each other. Its drawbacks (limited bandwidth, battery, multi-hop delay) are acceptable for short critical messages. Where a surviving base station or satellite link exists, it can be attached as a gateway to reach the outside network.

Cross-defence

- **Student A – why cellular depends on infrastructure:** nodes do not route; the base station gives radio access and channel control, and the core does authentication, switching, location tracking and handoff. Without towers, backhaul and power, no call can be set up.
- **Student B – how ad hoc continues without infrastructure:** each node acts as host and router, discovers neighbours by hello messages, builds routes through intermediate nodes, and when a link or node fails it repairs or rediscovers a route, so communication continues across whatever nodes remain.

Q4. Applications of Mobile IP and Routing Requirements

Shared work. Real-world applications that need mobility support include emergency communication, mobile healthcare, intelligent transportation, mobile banking and field-service systems, online gaming and video conferencing on the move, and military communication. We select **mobile healthcare (ambulance and patient monitoring)**.

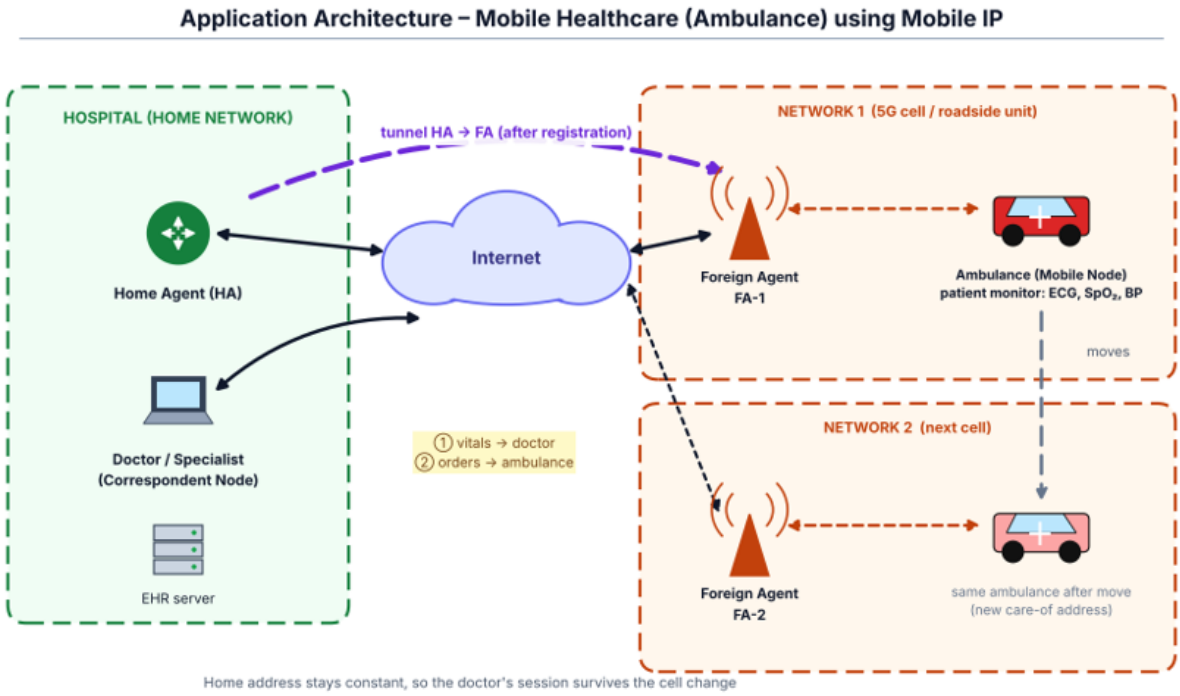
Mobile entities and requirements. Mobile entities: ambulances, paramedic tablets and patient-monitoring devices (ECG, SpO_2, blood pressure) that move across city areas. Fixed endpoints: the hospital server, doctor's workstation and electronic health record database. Requirements: continuous session while moving, low delay for vital-sign streams, high reliability, confidentiality of patient data and quick recovery after handoff.

Applying Mobile IP. Each ambulance device keeps a permanent home address registered with a Home Agent at the hospital network. When the ambulance enters another operator's or city zone it obtains a care-of address, registers with the HA, and the hospital server keeps sending to the same home address while the HA tunnels packets to the ambulance.

Routing requirements created by mobility. The network must handle changing next hops, fast route repair after handoff, minimal control overhead, loop-free routes and support for multiple access networks (4G/5G, Wi-Fi). Between ambulances in areas without coverage, an ad hoc routing protocol (such as AODV) can relay data to a node that has connectivity.

Evaluation of suitability. Mobile IP is suitable because the session must survive changes of network and the hospital knows a single address for each unit. Its weaknesses are registration delay, triangular routing and tunnelling overhead, which are reduced by route optimisation, fast handover and a Home Agent placed close to or inside the hospital network. Overall the solution

is suitable for non-instantaneous clinical data and acceptable for live telemetry when combined with these improvements.



Student B – Application architecture

- **Application:** mobile healthcare – live patient telemetry and teleconsultation from an ambulance.
- **Mobile nodes and endpoints:** mobile nodes are the ambulance gateway/tablet and sensors; endpoints (Correspondent Nodes) are the hospital server and the doctor's console.
- **Architecture:** Sensors → Ambulance gateway (MN) ⇄ cellular/Wi-Fi access network (FA / base station) ⇄ Internet ⇄ Hospital Home Agent and hospital server.

- **How Mobile IP supports mobility:** the home address stays constant; registration updates the HA binding at each new network; the HA tunnels packets to the current care-of address.
- **Required components:** Home Agent, Foreign Agent or DHCP for a co-located CoA, authentication server (AAA), secure VPN or IPsec tunnels, hospital server and database.
- **Communication flow:** (1) sensor data reaches the ambulance gateway; (2) the gateway sends it to the hospital server; (3) the doctor's replies go to the ambulance's home address; (4) the HA intercepts and tunnels them to the care-of address; (5) on a network change the gateway re-registers and the flow resumes.

Student A – Performance and requirement analysis

Aspect	Analysis
Communication delay	Direct uplink is short, but downlink via the HA adds the HA detour (triangular routing) and 20 bytes of tunnel header; target end-to-end delay for telemetry is below about 150–300 ms.
Packet delivery and connectivity	Delivery is good inside a cell; loss rises at cell edges and during handoff, so buffering and retransmission are needed for vital data.
Mobility requirements	Session continuity, fast handoff detection, a permanent address, and the ability to use more than one access network.

Interruption during movement	Gap between detachment and registration (about hundreds of milliseconds); mitigated by fast handover and buffering at the previous agent.
Security and reliability	Authenticated registration (HA–MN shared key), encrypted tunnels (IPsec), replay protection, HA redundancy, link redundancy and acknowledgements for critical messages.
Mobile IP suitability	Suitable: yes, with route optimisation, a nearby HA and secure tunnels; less suitable for hard real-time control where even brief interruption is unacceptable.

Q5. Proactive Routing Protocols in Mobile Networks (6 marks)

Shared work. Proactive (table-driven) protocols keep up-to-date routes to every destination at all times by exchanging routing information periodically and on topology change. A route is available immediately when needed, at the cost of continuous control traffic. Examples: DSDV, OLSR, WRP. We select **DSDV (Destination-Sequenced Distance Vector)**, a distance-vector protocol adapted for mobile nodes using destination sequence numbers.

Topology. Five mobile nodes A, B, C, D, E with links A–B, B–C, C–D, D–E (a chain), plus link B–D.

Routing table of A (before change)

Destination	Next hop	Metric (hops)	Seq. no.

A	A	0	A-100
B	B	1	B-100
C	B	2	C-100
D	B	2	D-100
E	B	3	E-100

Topology change: node D moves away and the links B–D, C–D and D–E break.

- B detects the broken link and sets the metric of D to ∞ with an odd sequence number D-101 (odd = generated by a neighbour for a broken route). B also invalidates E, which was reached through D, and sends a triggered update.
- A receives this and updates its table.

Routing table of A (after change)

Destination	Next hop	Metric (hops)	Seq. no.
A	A	0	A-102
B	B	1	B-102
C	B	2	C-102
D	B	∞	D-101
E	B	∞	E-101

When D reappears (say near C) it advertises a new even sequence number D-102 with metric 1, and every node replaces the ∞ entry because a higher sequence number always wins.

Routing under mobility. Tables stay current so any packet to a reachable destination is forwarded at once without discovery delay. The price is a stream of updates whenever nodes move, so overhead rises with mobility and network size.

Student A – Protocol design

- **Principle of proactive routing:** every node continuously maintains a route to every other node by periodic and triggered table exchanges, so routes are ready before they are needed.
- **DSDV operation:** each node keeps a table of destination, next hop, metric and sequence number. Nodes broadcast their full table (full dump, occasionally) or only changes (incremental update) to neighbours; a receiver adopts a route if it has a higher sequence number, or the same sequence number and a lower metric.
- **Routing-table maintenance:** periodic updates, plus triggered updates when a link breaks or a new route appears. Entries that stay stale beyond a timeout are removed. To avoid route fluctuation, DSDV delays advertising routes whose metric is still settling (settling time).
- **Sequence numbers:** each destination stamps its own advertisement with an even number that it increases; a broken-link route gets an odd number from the detecting node. They prevent routing loops (count-to-infinity) and give freshness: newer always beats older.
- **Sample topology:** shown above (A–B–C–D–E with extra link B–D).
- **Before and after:** the two tables above show that after D leaves, the entries for D and E become ∞ with higher sequence numbers.

ALGORITHM DSDVUpdate(Node, ReceivedTable, Sender)

INPUT: Node – receiving node; ReceivedTable – advertised routes; Sender – neighbour

```
OUTPUT: Updated routing table
BEGIN
  FOR each Route R in ReceivedTable DO
    NewMetric ← R.Metric + 1
    IF R.SequenceNumber > SequenceNumber[R.Destination] THEN
      UPDATE ROUTE R.Destination: NextHop ← Sender, HopCount ← NewMetric,
                                  SequenceNumber ← R.SequenceNumber
    ELSE IF R.SequenceNumber = SequenceNumber[R.Destination] AND NewMetric <
HopCount[R.Destination] THEN
      UPDATE ROUTE R.Destination: NextHop ← Sender, HopCount ← NewMetric
    END IF
  END FOR
  IF any route changed THEN SEND triggered update TO neighbours
  RETURN routing table
END
```

Student B – Performance investigation

Aspect	Analysis
Routing-control overhead	Periodic updates are sent even when nothing changes; a full dump grows with N (O(N) entries per node, $O(N^2)$ in the network), and each topology change adds triggered updates.

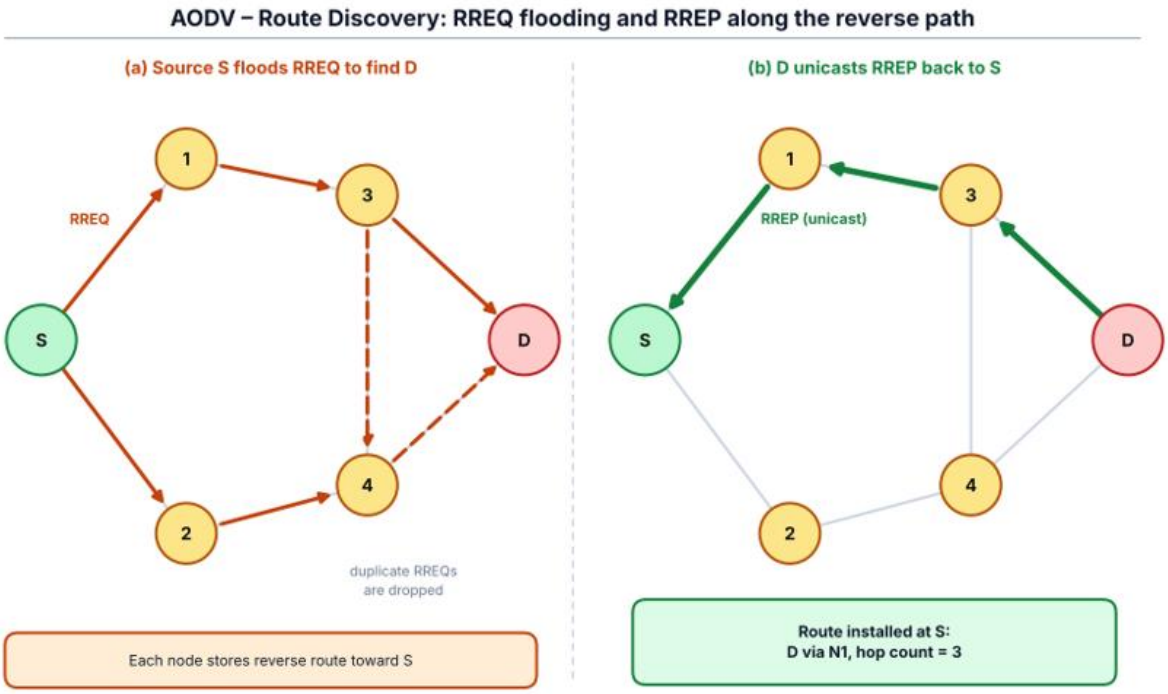
Route availability	Very high: a valid route to every known destination is always stored, so there is no route-discovery wait.
Convergence behaviour	Good in small or slow networks (a few update periods); sequence numbers prevent loops, but under fast movement tables are often stale and convergence lags the topology.
Control-message transmission	Broadcast to neighbours at fixed intervals plus triggered updates; incremental updates fit in one packet, full dumps need several.
Different network sizes	Small network (5–10 nodes): low overhead, fast convergence. Large network (50–100+ nodes): overhead and convergence time rise sharply and consume bandwidth and battery.
Delay and packet delivery	End-to-end delay is low because routes exist at once; delivery ratio is high when mobility is low, and falls when mobility is high because packets follow outdated next hops until updates arrive.

Q6. Reactive Routing and Protocol Comparison

Student A – Reactive routing investigation

- **Concept:** reactive (on-demand) routing finds a route only when a source has data to send, so no periodic table exchange is needed. Routes are cached while in use and removed when idle or broken.
- **AODV (Ad hoc On-demand Distance Vector):** each node keeps one next-hop entry per active destination with a destination sequence number, hop count and lifetime. It uses hop-by-hop forwarding and builds reverse and forward paths during discovery.
- **Route Request (RREQ):** when S needs a route to D, it broadcasts an RREQ (source, destination, broadcast ID, source and destination sequence numbers, hop count). Each receiver records a reverse route to S, drops duplicates, and rebroadcasts unless it knows a fresh route to D.
- **Route Reply (RREP):** D (or an intermediate node with a fresh route) unicasts an RREP back along the reverse path; each node on the way sets the forward route to D. S then starts sending data.
- **Route Error (RERR):** a node that detects a broken link (missed hello or failed transmission) invalidates affected routes, increments the destination sequence number and sends an RERR to the upstream nodes (precursors), which in turn invalidate their routes.
- **Route discovery and maintenance:** discovery = RREQ flood plus RREP; maintenance = local hello messages or link-layer feedback, RERR on failure, and rediscovery by the source if it still needs the route. Idle routes time out.

- **Effect of mobility:** more movement breaks active routes more often, so more RERRs and new discoveries; each discovery floods the network and delays the first packet. Routes not in use cost nothing, which keeps overhead lower than proactive protocols when traffic is light.



ALGORITHM AODVRouteDiscovery(Source, Destination)

INPUT: Source – originating node; Destination – target node

OUTPUT: Valid route or ROUTE_NOT_FOUND

BEGIN

IF RouteState[Destination] = VALID THEN RETURN NextHop[Destination]

SequenceNumber[Source] ← SequenceNumber[Source] + 1

BROADCAST RREQ(Source, Destination, BroadcastID, HopCount ← 0)

START TIMER discoveryTimeout

```
  WHILE NOT EXPIRED(discoveryTimeout) DO
    IF RREP received THEN
      ADD ROUTE Destination: NextHop ← RREP.Sender, HopCount ← RREP.HopCount + 1
      RouteState[Destination] ← VALID
      RETURN NextHop[Destination]
    END IF
  END WHILE
  RETURN ROUTE_NOT_FOUND
END
```

Student B – Comparative verification: AODV and DSDV

Parameter	DSDV (proactive)	AODV (reactive)
Route discovery	None; routes learnt through periodic and triggered table updates	On demand through RREQ flood and RREP
Routing-table requirements	Entry for every destination in the network; larger table	Entries only for active destinations; smaller table
Control overhead	Constant periodic overhead, grows with N and with mobility	Overhead only when routes are needed or break; low with light traffic, high with many new flows or breakage
Initial communication	Very low; route already exists	Higher; the first packet waits for route discovery (RREQ/RREP round trip)

delay		
Mobility handling	Link break advertised by triggered update; tables lag in fast movement	Break reported by RERR; the source rediscovers only if it still needs the route
Suitable scenarios	Small, fairly stable networks needing immediate delivery (conference hall, small office mesh)	Larger or highly dynamic networks with sparse traffic (disaster response, vehicles, sensor networks)

Justification of protocol choice for a given condition. Choose DSDV for a small, low-mobility network with constant traffic to many destinations, where immediate routes matter; choose AODV for a larger or more mobile network where most nodes talk to only a few destinations and bandwidth and battery are scarce.

Joint question

For a highly dynamic mobile network with frequent topology changes, would you select DSDV or AODV? Justify using routing overhead, delay and mobility considerations.

I would select **AODV**.

- **Routing overhead:** DSDV must send updates for every topology change to every node, and in a highly dynamic network that is a continuous flood of control packets (about $O(N^2)$ entries network-wide) that consumes bandwidth and battery. AODV generates control traffic only for routes that are in use, and only the affected routes react to a break.

- **Delay:** DSDV gives lower delay for the first packet because routes already exist, but under rapid change those routes are often stale, so packets are lost or misrouted until the next update. AODV pays a discovery delay (one RREQ/RREP round trip) but then uses a fresh route.
- **Mobility:** AODV's RERR plus rediscovery repairs only the broken routes and uses destination sequence numbers to guarantee loop-free, fresh routes, whereas DSDV's tables have to converge across the whole network after every move.

So AODV gives better delivery ratio and lower overhead in a highly mobile network, at the cost of some initial delay. If the application cannot tolerate the discovery delay, a hybrid protocol (such as ZRP) is a reasonable compromise.